Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 6 shows a device that removes small particles of soot from sooty air.

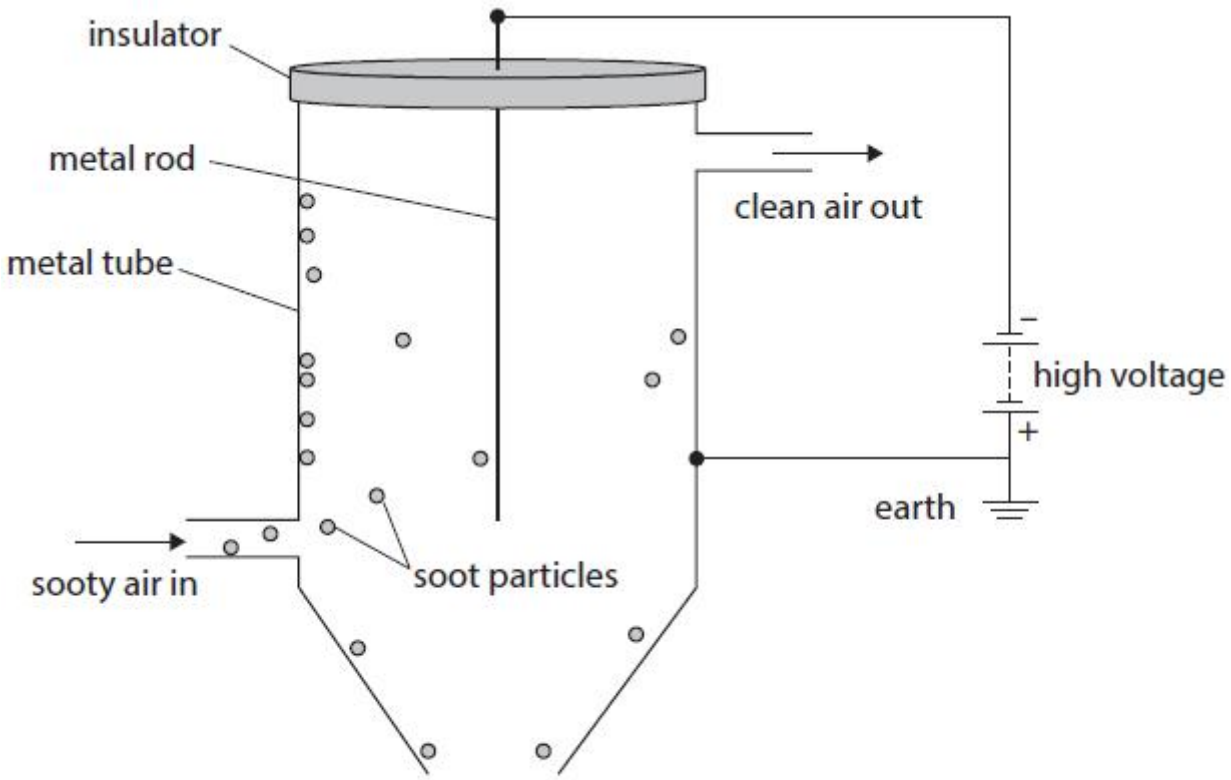

Figure 6

(i) There is a large potential difference between the metal tube and the metal rod in the centre of the tube.

This large potential difference causes the soot particles to gain a negative charge as they pass between the rod and the tube.
The soot particles have

(1)

- ☐ **A** lost protons
- ☐ **B** lost electrons
- ☐ **C** gained protons
- ☐ **D** gained electrons

(ii) Explain why, in Figure 6, the charged soot particles move towards the metal tube.

(2)

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(iii) When the soot particles reach the metal tube, the charge on them changes.

State how the charge on them changes.

(1)

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(iv) The metal tube does not become charged when the soot particles reach it.

Explain why the metal tube does not become charged.

(2)

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(Total for question = 6 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about static electricity.

A student has a rubber balloon tied to a long piece of cotton thread.

The student gives the balloon an overall electrostatic charge.

(i) Describe **one** way that the student could give the balloon an overall electrostatic charge.

(2)

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(ii) The student gives the balloon an overall negative charge.

Which of these sentences explains why the overall charge on the balloon is negative?

(1)

☐ **A** Negative charge has been removed from the balloon.
☐ **B** Negative charge has been added to the balloon.
☐ **C** Positive charge has been removed from the balloon.
☐ **D** Positive charge has been added to the balloon.

(iii) The student charges another balloon on a long thread.

Explain how the student can show that the two balloons have the same type of charge.

(3)

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(Total for question = 6 marks)

Q3.

A battery sends a current through a metal wire.

(a) (i) Complete the sentence by putting a cross (■) in the box next to your answer.

Direct current is movement of charge

(1)

☒ **A** backwards and forwards

☒ **B** in many directions

☒ **C** in one direction

☒ **D** up and down

(ii) Complete the sentence by putting a cross (■) in the box next to your answer.

The particles that flow in the metal wire are

(1)

☒ **A** atoms

☒ **B** electrons

☒ **C** protons

☒ **D** neutrons

(b) The current in a wire is 3.7 A.

Calculate the charge that flows into the wire in 13 s.

(2)

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(c) Plastic is an insulator.

A student rubs a piece of plastic with a cloth.

This gives the plastic a negative charge.
(i) Explain how the plastic is charged by the rubbing.

(2)

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(ii) The cloth is also charged when it rubs against the plastic.

Describe the charge on the cloth.

(2)

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(Total for Question is 8 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A student gives a plastic strip an overall electric charge.

(i) Describe **one** way that the student can give the plastic strip an overall electric charge.

(1)

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(ii) Figure 1 shows a gold leaf electroscope that can be used to investigate static electricity.

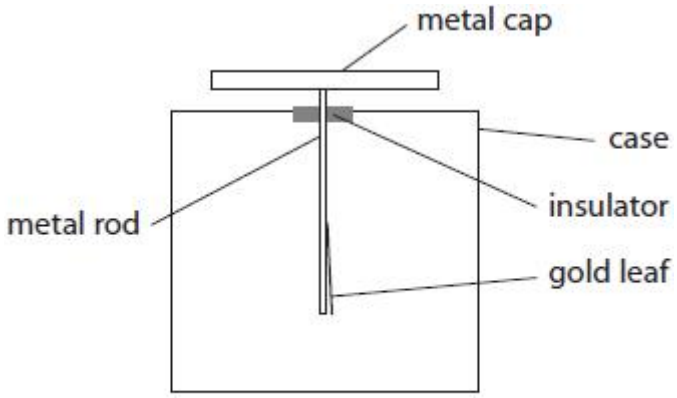
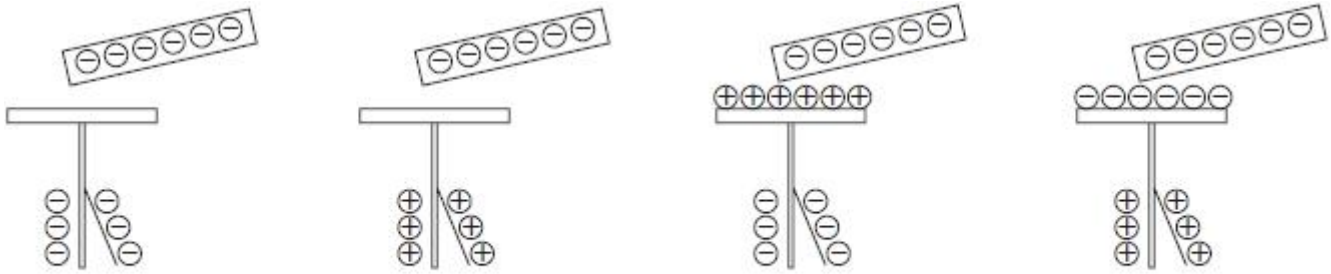


Figure 1

The electroscope has no overall charge.
The gold leaf has a very small mass and can bend very easily.
The student brings a negatively charged plastic strip near to the cap of the electroscope.
The gold leaf bends away from the metal rod.
Which diagram shows the way that electric charge is now distributed?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(Total for question = 2 marks)

Q5.

Figure 1 shows a prize that is made from a metal star on a plastic base.

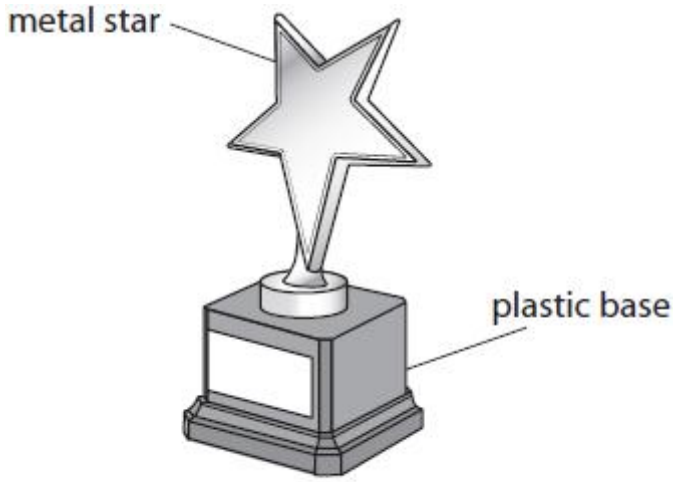


Figure 1

The charged plastic base attracts some dust from the air.

Figure 2 shows a magnified view of part of the surface of the plastic base and a dust particle.

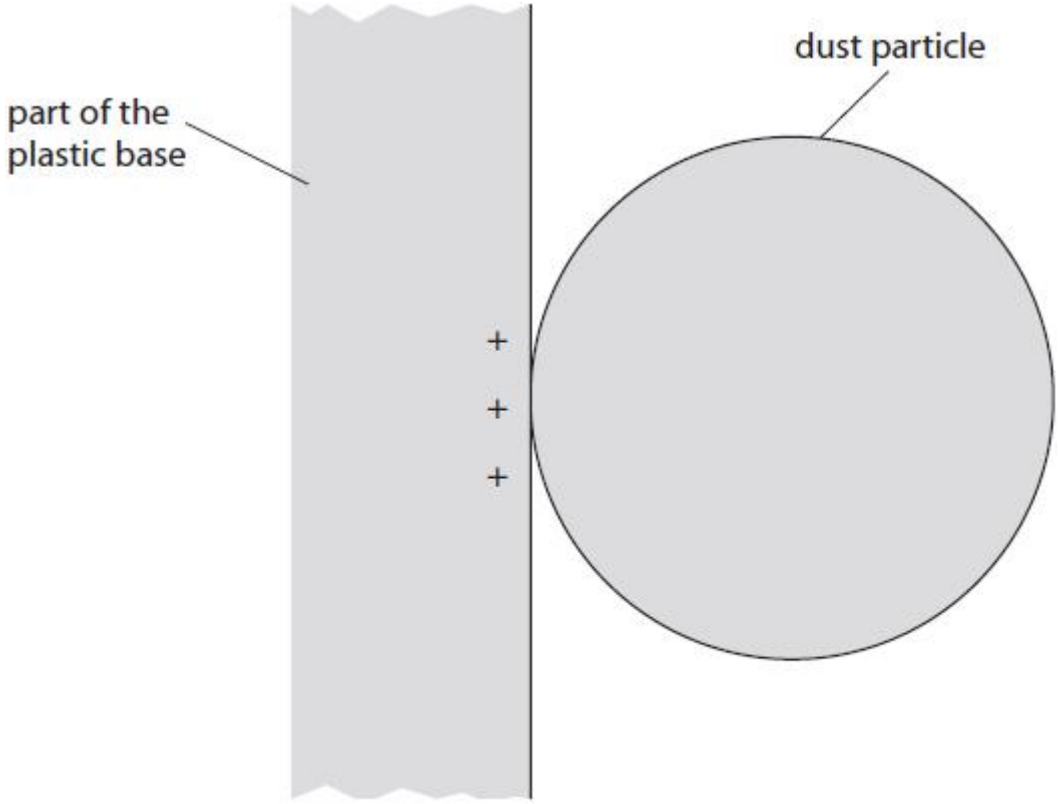


Figure 2

Some of the charges on the plastic base are shown but the charges induced on the dust particle are not shown.

Draw the charges induced on the dust particle in Figure 2.

(Total for question = 2 marks)

Q6.

Figure 3 shows two charged metal plates.

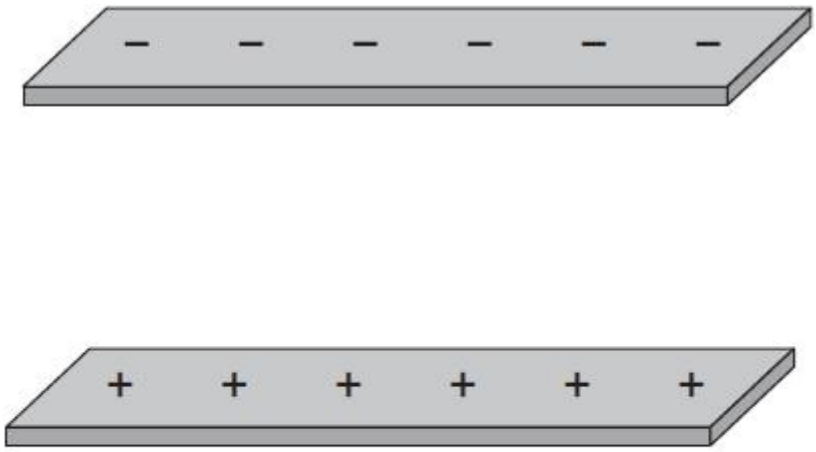


Figure 3

The top plate has a negative electric charge.

The bottom plate has a positive electric charge.

On Figure 3, draw the electric field lines between the two plates and show the direction of this electric field.

(Total for question = 2 marks)

Q7.

This question is about static electricity.

Draw on Figure 1 the shape and direction of the electric field due to the positive point charge.

(2)

Figure 1

(Total for question = 2 marks)

Q8.

Figure 5 shows **two** metal plates.

The plates are charged by a very high voltage (potential difference).

Draw, on Figure 5, the shape and direction of the electric field between the two plates.

(2)

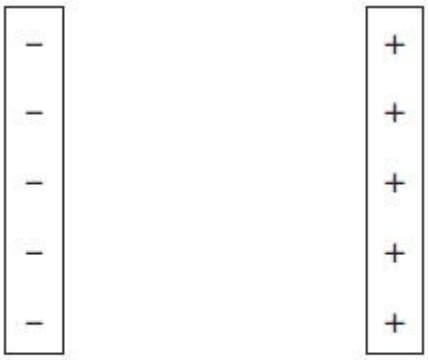


Figure 5

(Total for question = 2 marks)

Q9.

Two small objects, P and Q, are each given an electric charge.

Figure 3 represents the electric fields around the objects, P and Q.

Figure 3

(i) Use information from Figure 3 to give **two** differences between the charge on P and the charge on Q.

(2)

1 .....

.....

2 .....

.....

(ii) Object P and object Q are held near to each other so that their electric fields interact with each other. State the effect that the electric field of object Q has on object P.

(1)

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.....

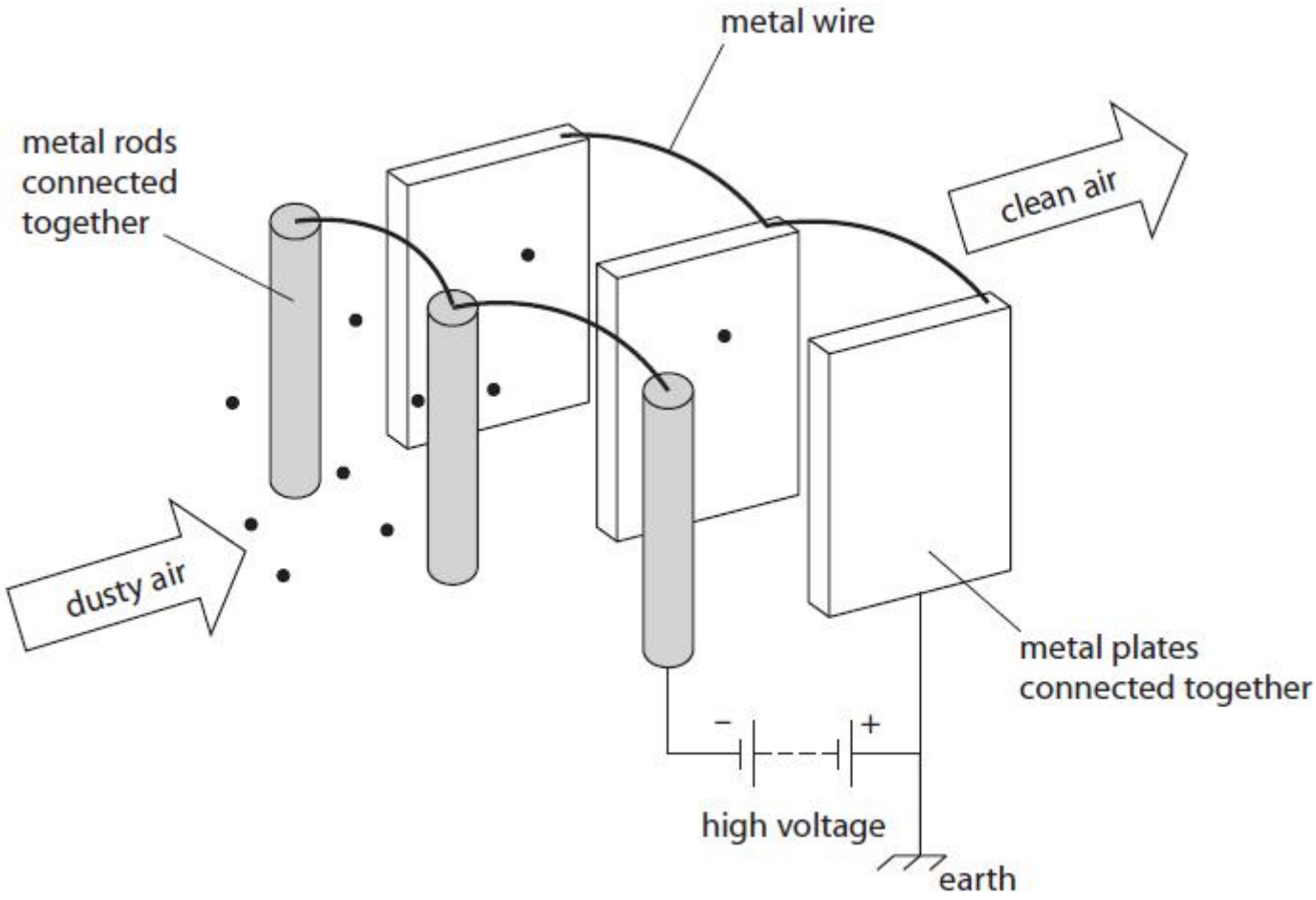
(Total for question = 3 marks)

Q10.

An electrostatic air filter is designed to remove dust particles from the air in a room.

A fan blows dusty air past several metal rods and metal plates.

There is a large potential difference (voltage) between the metal rods and the metal plates.



(a) Complete the sentence by putting a cross (☒) in the box next to your answer.

When dusty air goes past the metal rods, the dust particles become negatively charged.
This is because the dust particles

(1)

☐ **A** lose electrons

☐ **B** lose protons

☐ **C** gain electrons

☐ **D** gain protons

(b) When the dusty air flows past the metal plates, the dust particles settle on the metal plates.

Explain why the dust particles settle on the metal plates.

(2)

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(c) (i) State what happens to the charge on the dust particles when they settle on the metal plates.

(1)

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(ii) Explain why the charge does not build up on the metal plates.

(2)

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(d) There is a current of 1.2 mA in the circuit.

Calculate the charge transferred by this current in 40 s.
State the unit.

(3)

charge transferred = ... unit: .....

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	D gained electrons A and C are incorrect because protons are not exchanged in this context B is incorrect because it would give the particles a net positive charge		(1) AO1.1

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking any **two** from (soot particles are in an) electric field (1) **AND EITHER** attraction (1) (surface of) it/tube has (induced) positive/opposite/unlike charge (1) **OR** repulsion (1) rod (and soot particles) has a negative/like/same charge (1)		(2) AO1.1

Question number	Answer	Additional guidance	Mark
(iii)	(net charge) becomes zero (1)	discharged / charge is lost / charge moves on to the metal tube/ becomes neutral allow electrons for charge ignore becomes positively charged	**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
(iv)	an explanation linking metal is a conductor (1) (connected to) earth/ground (1)	allow electrons for charge charge moves through the metal / tube / wire	**(2)** **AO1**

Q2.

Question number	Answer	Additional guidance	Mark
(i)	rub (the balloon) (1) with a piece of cloth/hair/fur (1)	use friction allow on any insulated object	**(2)**

Question number	Answer	Additional guidance	Mark
(ii)	B Negative charge has been added to the balloon A is incorrect removing negative charge would make the balloon positively charged. C and D are incorrect because positive charge cannot be moved		(1)

Question number	Answer	Additional guidance	Mark
(iii)	an explanation linking: method of handling balloons without discharging them (1) bring balloons near to each other (1) observation of repulsion (1)	 hang balloons up by their strings they/balloons will push away (from each other)	(3)

Q3.

	Answer	Acceptable answers	Mark
(a)(i)	C		(1)
(a)(ii)	B		(1)
(b)	substitution (1) 3.7 x 13 evaluation (1) 48 (C)	48.1 Correct answer with no calculation scores 2 marks	(2)
(c)(i)	Correct responses can be seen in (i) or (ii) An explanation linking • <u>electrons</u> (1)an	["positive electrons/ protons moving", seen anywhere in part (i) or (ii) loses this mark] ignore reference to charge	(2)

	d <u>one</u> of • removed by friction (1) • (transferred) <u>to</u> plastic (1)	before rubbing transferred from cloth	
(c)(ii)	opposite to charge on plastic (1) <u>equal</u> to charge on the plastic (1)	charge on cloth is positive <u>same size</u> as charge on plastic electrons transferred from the cloth equal to electrons lost by cloth	**(2)**

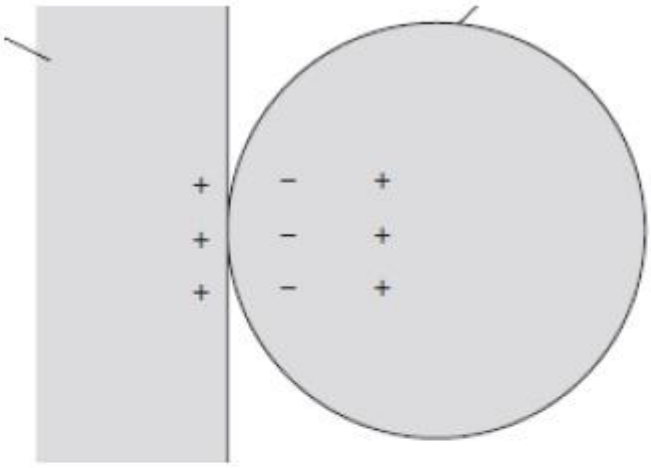
Total question = 8 marks

Q4.

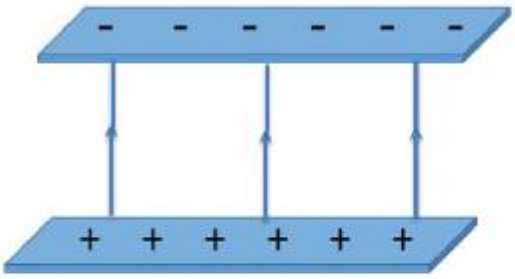
Question number	Answer	Additional guidance	Mark
(i)	use friction (1)	rub (the plastic rod)	**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
(ii)	C A and B are incorrect because the cap would become charged D is incorrect because the cap would have an opposite charge to that on the rod		**(1)** **AO1**

Q5.

Question number	Answer	Additional guidance	Mark
	 at least one negative charge on left hand side of dust particle (1) at least one positive charge to the right of the negative charge(s) (1) net negative charge on left hand side / near to base (1) net positive charge to the right of the negative charge / further from base than negative charge (1)	ignore charges drawn on base	**(2)** **AO1**

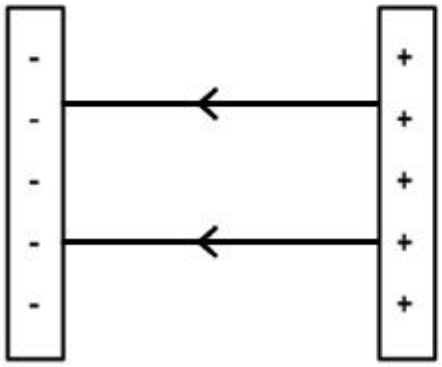
Q6.

Question Number	Answer	Additional Guidance	Mark
	 at least three vertical straight lines (equally spaced) (1) **with** at least one arrow in the right direction (1)	judge by eye any arrow in the wrong direction = 1 mark max	**(2)**

Q7.

Question number	Answer	Additional guidance	Mark
	at least three **radial** lines from the charge (1) direction shown away from the charge (1)	do not allow curved lines ignore circles without arrows consistently	**(2)** **AO1.2**

Q8.

Question number	Answer	Additional guidance	Mark
	at least two horizontal straight lines between the two plates, judge by eye (1) arrows **only** from +ve towards -ve (1)	independent marks lines do not have to touch plates ignore curved lines at top and bottom of plate(s) example, scores 2 marks 	**(2)** **AO1.1**

Q9.

Question number	Answer	Additional guidance	Mark
(i)	(size of) charge on Q is greater than (size of charge) on P (1) P has (overall) negative charge and Q has (overall) positive charge (1)	in any order Q has more charge / stronger field than P accept abbreviations such as +'ve, -'ve charge on P is opposite to charge on Q	**(2)** **AO3**

Question number	Answer	Additional guidance	Mark
(ii)	(force of) attraction on (object) P from (object) Q (1)		(1) AO1

Q10.

Question Number	Answer	Acceptable answers	Mark
(a)	C (gain electrons)		(1)

Question Number	Answer	Acceptable answers	Mark
(b)	An explanation linking • (Force of) attraction (1) • (plates have) opposite charge (to dust) (1)	Plates have a positive charge Ignore different charge	(2)

Question Number	Answer	Acceptable answers	Mark
(c)(i)	transferred to plate / lost (1)	neutral / become discharged	(1)

Question Number	Answer	Acceptable answers	Mark
(c)(ii)	An explanation linking any two of • Metal is a conductor (1) • Electrons / (negative) charge moves (through the plates/ wire) (1) • Towards the voltage supply / earth /ground (1)	Metal not an insulator Plates / charges are earthed	(2)

Question Number	Answer	Acceptable answers	Mark
(d)	Substitution: $Q = 1.2 \times 10^{-3} \times 40$ (1) Evaluation: 0.048 or 4.8×10^{-2} (1) C / coulombs (1)	Give 2 marks for correct answer with no working shown Unit mark is independent Allow for 1 mark 48 (with incorrect or no units) Allow for 2 marks 48 C Allow for all 3 marks 48 mC	**(3)**